In other words, there are two distinct eigenvectors with the same eigenvalue. It should be clear that any linear combination of the two eigenvectors is also an eigenvector with the same eigenvalue. With this much freedom, it is always possible to find two orthogonal linear combinations.

Let's see how. Consider an arbitrary linear combination of these two eigenvectors:

$$|A\rangle = \alpha|\lambda_1\rangle + \beta|\lambda_2\rangle.$$

Operating on both sides with **L**, we get

$$\mathbf{L}|A\rangle = \alpha\mathbf{L}|\lambda_1\rangle + \beta\mathbf{L}|\lambda_2\rangle,$$

$$\mathbf{L}|\mathrm{A}\rangle = \alpha\lambda|\lambda_1\rangle + \beta\lambda|\lambda_2\rangle,$$

and finally

$$\mathbf{L}|\mathrm{A}\rangle = \lambda(\alpha|\lambda_1\rangle + \beta|\lambda_2\rangle) = \lambda|\mathrm{A}\rangle.$$

This equation demonstrates that any linear combination of $|\lambda_1\rangle$ and $|\lambda_2\rangle$ is also an eigenvector of **L**, with the same eigenvalue. By assumption, these two vectors are linearly independent—otherwise, they would not represent distinct states. We will also suppose that they span the subspace of eigenvectors of **L** that have eigenvalue λ. There is a straightforward process, called the *Gram-Schmidt* procedure, for finding an orthonormal basis for a subspace, given a set of independent vectors that spans the subspace. In plain English, we can find two orthonormal eigenvectors by writing them as a linear combination of $|\lambda_1\rangle$ and $|\lambda_2\rangle$. We outline the Gram-Schmidt procedure below, in Section 3.1.6.